RACE Regular Analysis Through Continuous Exercise

Modern Physics : RACE-01

1. The stopping potential in the context of photoelectric effect depends on the following property of incident electromagnetic radiation:

(A) Phase (B) Intensity (C) Amplitude (D) Frequency

PMP261

2. If 2.5×10^{-6} N average force is exerted by a light wave on a non-reflecting surface of 30 cm^2 area during 40 minutes of time span, the energy flux of light just before it falls on the surface is ___ W/cm^2.

(Round off to the Nearest Integer)

(Assume complete absorption and normal incidence conditions are there)

PMP262

3. Conductivity of a photodiode starts changing only if the wavelength of incident light is less than 660 nm. The band gap of photodiode is found to be $\left(\frac{X}{8}\right)eV$. The value of X is :

(Given, $h = 6.6 \times 10^{-34}$ Js, $e = 1.6 \times 10^{-19}C$)

(A) 15 (B) 11 (C) 13 (D) 21

PMP263

4. Monochromatic light of frequency 6×10^{14} Hz is produced by a laser. The power emitted is 2×10^{-3} W. How many photons per second on an average, are emitted by the source ?

(Given $h = 6.63 \times 10^{-34}$ Js)

(A) 9×10^{18} (B) 6×10^{15} (C) 5×10^{15} (D) 7×10^{16}

PMP264

5. An object is placed in a medium of refractive index 3. An electromagnetic wave of intensity 6×10^8 W/m^2 falls normally on the object and it is absorbed completely. The radiation pressure on the object would be (speed of light in free space = 3×10^8 m/s) :

(A) 36 Nm^{-2} (B) 18 Nm^{-2} (C) 6 Nm^{-2} (D) 2 Nm^{-2}

PMP265

6. Two sources of light emit with a power of 200 W. The ratio of number of photons of visible light emitted by each source having wavelengths 300 nm and 500 nm respectively, will be :

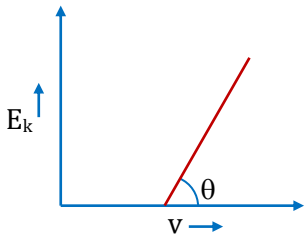
(A) 1 : 5 (B) 1 : 3 (C) 5 : 3 (D) 3 : 5

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7. For the photoelectric effect, the maximum kinetic energy (E_k) of the photoelectrons is plotted against the frequency (v) of the incident photons as shown in figure. The slope of the graph gives



(A) Ratio of Planck's constant to electric charge

(B) Work function of the metal

(C) Charge of electron

(D) Planck's constant

PMP267

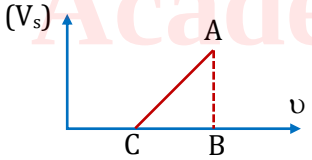
8. Threshold wavelength of tungsten is 2300 angstrom. If ultraviolet light of wavelength 1800 angstrom is incident on it, then the maximum kinetic energy of photoelectrons would be ?

(A) 1.5 eV (B) 2.5eV (C) 3.0 eV (D) 5.0eV

PMP268

9. Graph between stopping potential for most energetic emitted photoelectrons (V_s) with frequency (v) of incident radiation on metal is given below. Value of AB/BC, in graph is [where h = plank's constant, e = electronic charge]

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(A) h (B) e (C) h/e (D) e/h

PMP269

10. When a metal surface is illuminated by light of wavelengths 400 nm and 250 nm, the maximum velocities of the photoelectlons ejected are v and $2v$ respectively. The work function of the metal is (h : Planck's constant, c : velocity of light in air)

(A) $2hc\times10^6 J$ (B) $1.5hc\times10^6 J$ (C) $hc\times10^6 J$ (D) $0.5hc\times10^6 J$

PMP270

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	25	A	C	C	D	D	A	C	A

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RACE Regular Analysis Through Continuous Exercise

Modern Physics : RACE-02

1. An electron of mass m and a photon have same energy E. The ratio of wavelength of electron to that of photon is : (c being the velocity of light)

(A) $\frac{1}{c}\left(\frac{2m}{E}\right)^{1/2}$ (B) $\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$ (C) $\left(\frac{E}{2m}\right)^{1/2}$ (D) $c\,(2mE)^{1/2}$

PMP271

2. The wavelength of an X-ray beam is 10Å. The mass of a fictitious particle having the same energy as that of the X-ray photons is $\frac{x}{3}\,h\,kg$. The value of x is_____.

(h = Planck's constant)

PMP272

3. An α particle and a proton are accelerated from rest by a potential difference of 200 V. After this, their de Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$ is:

(A) 3.8 (B) 8 (C) 7.8 (D) 2.8

PMP273

4. An electron of mass m and magnitude of charge $|e|$ initially at rest gets accelerated by a constant electric field E. The rate of change of de-Broglie wavelength of this electron at time t ignoring relativistic effects is :

(A) $\frac{-h}{|e|Et^2}$ (B) $\frac{|e|Et}{h}$ (C) $-\frac{h}{|e|E\sqrt{t}}$ (D) $-\frac{h}{|e|Et}$

PMP274

5. A particle A of mass m and initial velocity v collides with a particle B of mass $\frac{m}{2}$ which is at rest. The collision is head on, and elastic. The ratio of the de–Broglie wavelengths λ_A to λ_B after the collision is :

(A) $\frac{\lambda_A}{\lambda_B}=\frac{2}{3}$ (B) $\frac{\lambda_A}{\lambda_B}=\frac{1}{2}$ (C) $\frac{\lambda_A}{\lambda_B}=\frac{1}{3}$ (D) $\frac{\lambda_A}{\lambda_B}=2$

PMP275

6. The de-Broglie wavelength associated with the electron in the n = 4 level is :-

(A) two times the de-Broglie wavelength of the electron in the ground state

(B) 1/4th of the de-Broglie wavelength of the electron in the ground state

(C) four times the de-Broglie wavelength of the electron in the ground state

(D) half of the de-Broglie wavelength of the electron in the ground state

PMP276

7. The de Broglie wavelengths of a proton and an α particle are λ and 2 λ respectively. The ratio of the velocities of proton and α particle will be :

(A) 1 : 8 (B) 1 : 2 (C) 4 : 1 (D) 8 : 1

PMP277

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8. The de-Broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of K.E. of electron and K.E. of photon will be:

(A) $\frac{1}{1}$ (B) $\frac{1}{8}$ (C) $\frac{8}{1}$ (D) $\frac{1}{4}$

PMP278

9. The waves emitted when a metal target is bombarded with high energy electrons are

(A) Microwaves (B) *X*-rays (C) Infrared rays (D) Radio Waves

PMP279

10. During the transition of electron from state *AS* to state *C* of a Bohr atom, the wavelength of emitted radiation is 2000 Å and it becomes 6000 Å when the electron jumps from state *B* to state *C*. Then the wavelength of the radiation emitted during the transition of electrons from state *A* to state *B* is:

(A) 3000 Å (B) 6000 Å (C) 4000 Å (D) 2000 Å

PMP280

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	10	D	A	D	C	D	B	B	A

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RACE Regular Analysis Through Continuous Exercise

Modern Physics : RACE-03

1. A particular hydrogen - like ion emits the radiation of frequency 3×10^{15} *Hz* when it makes transition from $n = 2$ to $n = 1$. The frequency of radiation emitted in transition from $n = 3$ to $n = 1$ is $\frac{x}{9} \times 10^{15}$ *Hz*, when $x =$ _____.

PMP281

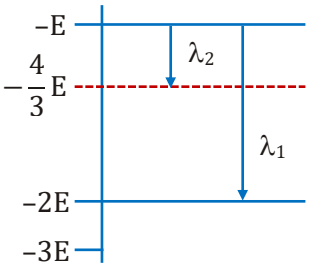
2. Match List-I (Fundament Experiment) with List-II (its conclusion) and select the correct option from the choices given below the list :

List-I		List-II	
(A)	Franck-Hertz Experiment	(i)	Particle nature of light
(B)	Photo-electric experiment	(ii)	Discrete energy levels of atom
(C)	Davison-Germer Experiment	(iii)	Wave nature of electron
		(iv)	Structure of atom

(A) A-ii, B-i, C-iii (B) A-iv, B-iii, C-ii (C) A-i, B-iv, C-iii (D) A-ii, B-iv, C-iii

PMP282

3. Some energy levels of a molecule are shown in the figure. The ratio of the wavelengths $r = \lambda_1/\lambda_2$, is given by :



(A) $r = \frac{3}{4}$ (B) $r = \frac{1}{3}$ (C) $r = \frac{4}{3}$ (D) $r = \frac{2}{3}$

PMP283

4. The time period of revolution of electron in its ground state orbit in a hydrogen atom is 1.6×10^{-16} *s*. The frequency of revolution of the electron in its first excited state (in s^{-1}) is:

(A) 6.2×10^{15} (B) 5.6×10^{12} (C) 7.8×10^{14} (D) 1.6×10^{14}

PMP284

5. The recoil speed of a hydrogen atom after it emits a photon in going from $n = 5$ state to $n = 1$ state will be :-

(A) 4.17 *m/s* (B) 2.19 *m/s* (C) 3.25 *m/s* (D) 4.34 *m/s*

PMP285

6. A particle of mass m moves in a circular orbit in a central potential field $U(r) = U_0 r^4$. If Bohr's quantization conditions are applied, radii of possible orbitals r_n vary with $n^{1/\alpha}$, where α is_____.

PMP286

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7. Choose the correct option from the following options given below :
(A) In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's model electrons always experience a net-force.
(B) An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-uniform mass distribution in Thomson's model
(C) A classical atom based on Rutherford's model is doomed to collapse.
(D) The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

PMP287

8. An electron revolving in n^{th} Bohr orbit has magnetic moment μ_n. If $\mu_n \alpha n^x$, the value of x is:
(A) 2 (B) 1 (C) 3 (D) 0

PMP288

9. The ratio of the magnitude of the kinetic energy to the potential energy of an electron in the 5th excited state of a hydrogen atom is :
(A) 4 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1

PMP289

10. At a given instant, say $t = 0$, two radioactive substances A and B have equal activities. The ratio $\frac{R_B}{R_A}$ of their activities after time t itself decays with time t as e^{-3t}. If the half-life of A is $\ell n2$, the half-life of B is :
(A) $\frac{\ell n2}{2}$ (B) $2ln2$ (C) $\frac{\ell n2}{4}$ (D) $4\ell n2$

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PMP290

11. A radioactive nucleus A with a half life T, decays into a nucleus B. At $t = 0$, there is no nucleus B. At sometime t, the ratio of the number of B to that of A is 0.3. Then, t is given by :
(A) $t = T \log (1.3)$ (B) $t = \frac{T}{\log(1.3)}$ (C) $t = \frac{T}{2}\frac{\log 2}{\log 1.3}$ (D) $t = T\frac{\log 1.3}{\log 2}$

PMP291

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	32	A	B	C	A	3	C	B	C	C
Question	11									
Answer	D									

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RACE Regular Analysis Through Continuous Exercise

Modern Physics : RACE-04

1. Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 *MeV*. It splits in two fragments 'B' and 'C' of mass numbers 105 and 115. The binding energy of nucleons in 'B' and 'C' is 6.4 *MeV* per nucleon. The energy Q released per fission will be :

(A) 0.8 *MeV* (B) 275 *MeV* (C) 220 *MeV* (D) 176 *MeV*

PMP292

2. A nucleus with mass number 184 initially at rest emits an α-particle. If the Q value of the reaction is 5.5 *MeV*, calculate the kinetic energy of the α-particle.

(A) 5.0 *MeV* (B) 5.5 *MeV* (C) 0.12 *MeV* (D) 5.38 *MeV*

PMP293

3. A nucleus of mass M emits γ-ray photon of frequency 'ν'. The loss of internal energy by the nucleus is :

[Take 'c' as the speed of electromagnetic wave]

(A) $h\nu$ (B) 0 (C) $h\nu\left[1-\frac{h\nu}{2Mc^2}\right]$ (D) $h\nu\left[1+\frac{h\nu}{2Mc^2}\right]$

PMP294

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4. The decay of a proton to neutron is :

(A) not possible as proton mass is less than the neutron mass

(B) possible only inside the nucleus

(C) not possible but neutron to proton conversion is possible

(D) always possible as it is associated only with β^+ decay

PMP295

5. A radioactive sample is undergoing α decay. At any time t_1, its activity is A and another time t_2, the activity is $\frac{A}{5}$. What is the average life time for the sample ?

(A) $\frac{\ell n5}{t_2-t_1}$ (B) $\frac{t_1-t_2}{\ell n5}$ (C) $\frac{t_2-t_1}{\ell n5}$ (D) $\frac{\ell n(t_2+t_1)}{2}$

PMP296

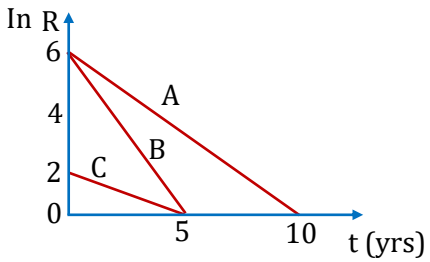
6. Two radioactive substances X and Y originally have N_1 and N_2 nuclei respectively. Half life of X is half of the half life of Y. After three half lives of Y, number of nuclei of both are equal. The ratio $\frac{N_1}{N_2}$ will be equal to :

(A) $\frac{1}{8}$ (B) $\frac{3}{1}$ (C) $\frac{8}{1}$ (D) $\frac{1}{3}$

PMP297

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7. Activities of three radioactive substances A, B and C are represented by the curves A, B and C, in the figure. Then their half-lives $T_{\frac{1}{2}}(A):T_{\frac{1}{2}}(B):T_{\frac{1}{2}}(C)$ are in the ratio :



(A) 3 : 2 : 1 (B) 4 : 3 : 1 (C) 2 : 1 : 3 (D) 2 : 1 : 1

PMP298

8. The activity of a radioactive sample falls from 700 s^{-1} to 500 s^{-1} in 30 minutes. Its half life is close to:

(A) 66 min (B) 52 min (C) 72 min (D) 62 min

PMP299

9. Using a nuclear counter the count rate of emitted particles from a radioactive source is measured. At t = 0 it was 1600 counts per second and t = 8 seconds it was 100 counts per second. The count rate observed, as counts per second, at t = 6 seconds is close to :

(A) 150 (B) 360 (C) 200 (D) 400

PMP300

10. Consider the nuclear fission

$Ne^{20} \rightarrow 2He^4 + C^{12}$

Given that the binding energy/nucleon of Ne^{20}, He^4 and C^{12} are, respectively, 8.03 *MeV*, 7.07 *MeV* and 7.86 *MeV*, identify the correct statement :

(A) 8.3 *MeV* energy will be released

(B) energy of 12.4 *MeV* will be supplied

(C) energy of 11.9 *MeV* has to be supplied

(D) energy of 3.6 *MeV* will be released

PMP301

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	D	B	C	C	C	D	C	C

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